Lab 9: Configure Virtual Private Clouds (VPCs) for secured data access

A Virtual Private Cloud (VPC) enables users to create a logically isolated section containing a pool of resources in a public cloud. Users can specify the public/private subnets and set inbound & outbound rules of the traffic in the Firewall/Security Groups for secure data access.

This lab session will let students explore 1) how to design a VPC and its subnets to create a logically isolated network environment in a public cloud, and 2) how to develop a web application with a Relational Database Service (RDS) and an Elastic Compute Cloud (EC2) in the VPC subnets for safer database access.

Topics covered

Students need to complete the design and development of the web application by using Amazon VPCs, subnets, security group, RDS and EC2 services. A high-level system architecture is as shown in Figure 1 below.

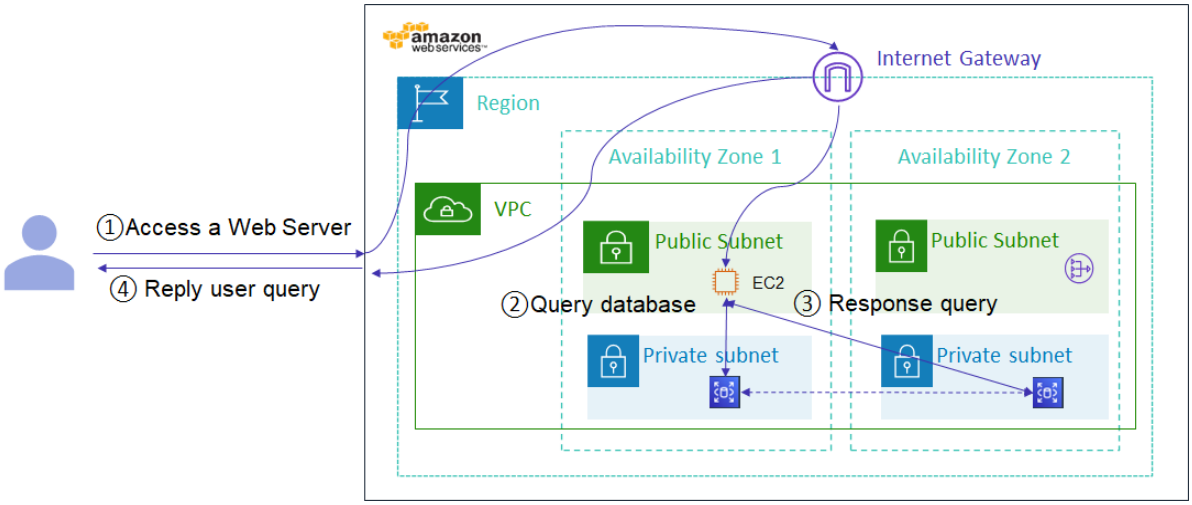


Figure 1: Create a Simple Web Application to access RDS database in VPC subnets

Learning Objectives

- Launching a VPC with private and public subnets
- Creating the Security Group for VPC
- Creating a private subnet group to contain the RDS database
- Launching an Amazon EC2 Web Server instance.
- Launching an Amazon Database Service (RDS) instance
- Developing a simple web application for recording the attendance

Duration

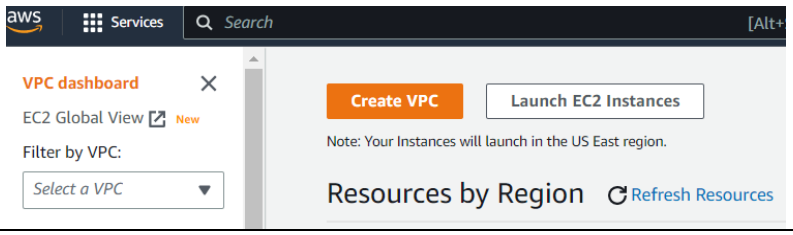
This lab takes approximately **100 minutes** to complete.

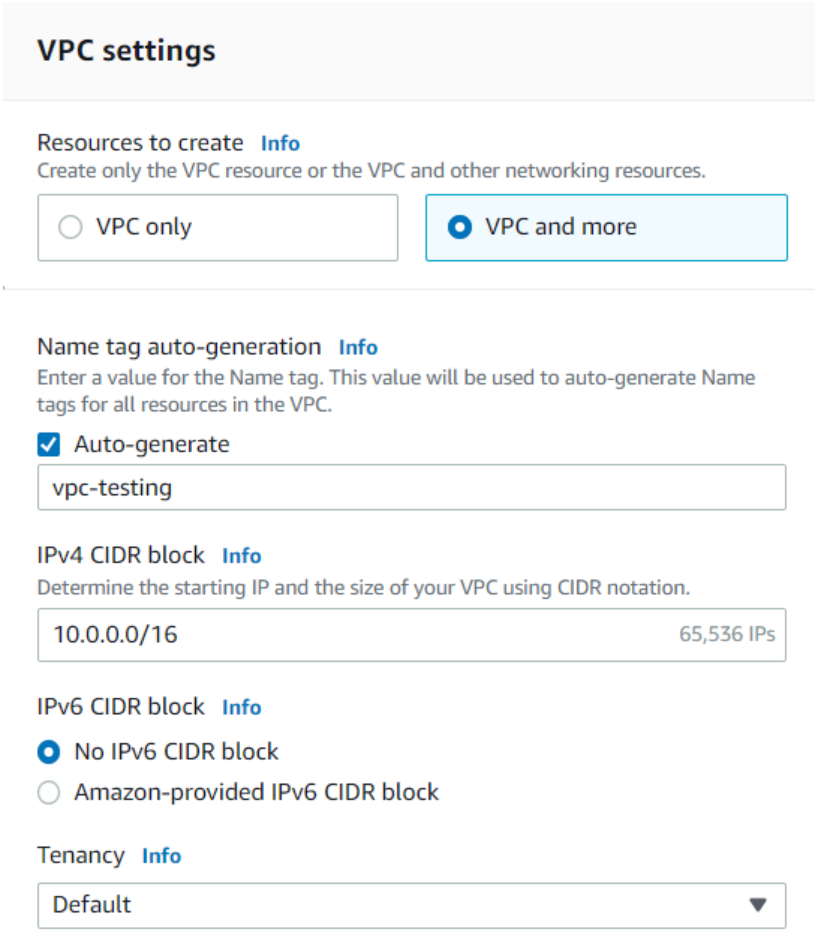
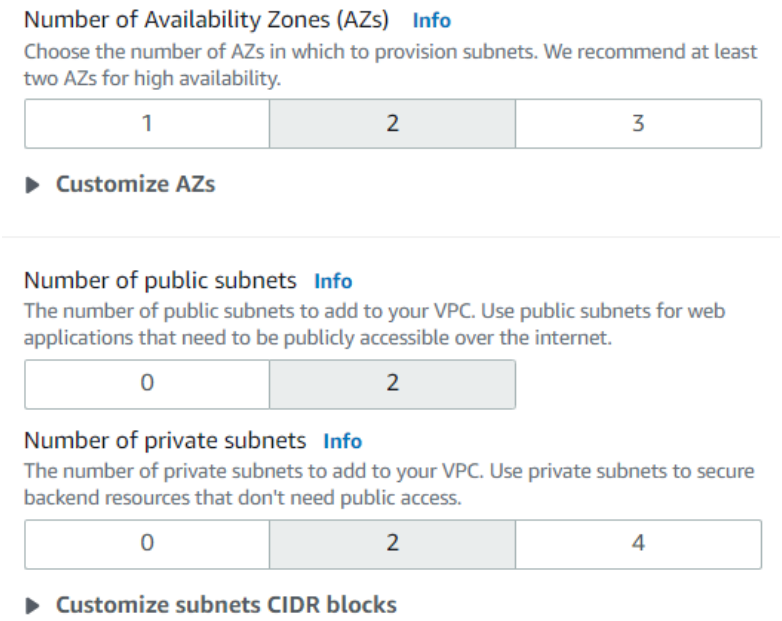
Build a Web Application for Student Attendance Records

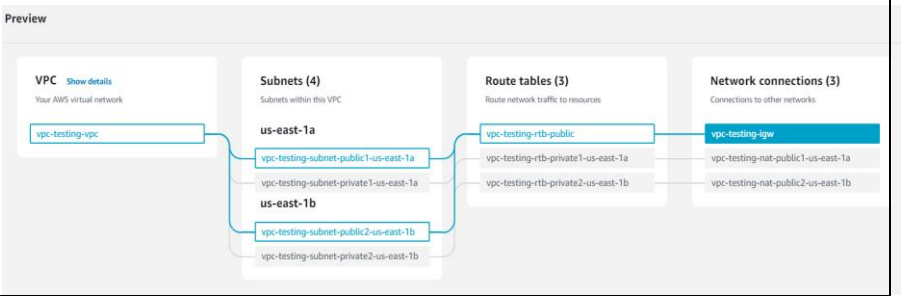
A Virtual Private Cloud (VPC) is a logically isolated environment where users can configure the secure access to the cloud resources in a public cloud. Students will need to create a VPC with its public and private subnets, configure the security group, and launch a Database Service (RDS) instance and an EC2 instance with the Apache web server with PHP installed for deploying a web application to record the attendance of a class.

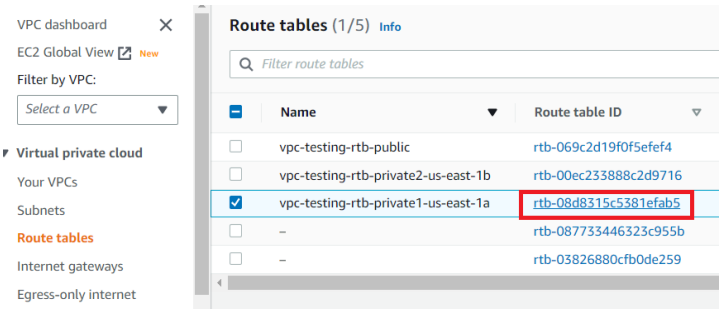
Task 1: Create a VPC with subnets and the security group

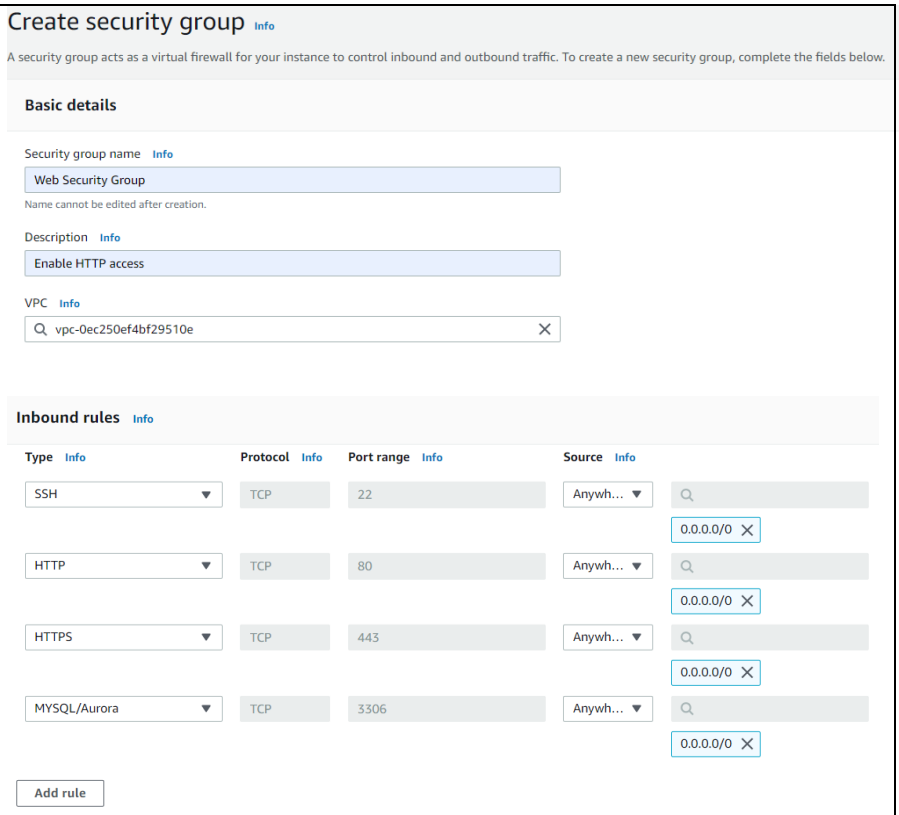
1. Search for VPC in the **Services** search box and choose **VPC** to open the VPC console.

SN	Steps	Instructions
1)	Create a VPC	Choose the **VPC dashboard,** then click **Create VPC** 

2)	Name VPC: **vpc-testing**	Configure the VPC settings: Choose VPC and more, name the VPC as vpc-testing, keep **IPv4 CIDR block** as 10.0.0.0/16, and leave the following using the default values. 
3)	Configure subnets: **2 AZs;** **2 public subnets;** **2 private subnets**	

		Check the CIDR blocks of the subnets by clicking Customize subnets CIDR blocks and **note down** the CIDR codes for subnets. Use **the default CIDR blocks** for the subnets. ▼ **Customize subnets CIDR blocks** Public subnet CIDR block in us-east-1a 10.0.0.0/20 — 4,096 IPs Public subnet CIDR block in us-east-1b 10.0.16.0/20 — 4,096 IPs Private subnet CIDR block in us-east-1a 10.0.128.0/20 — 4,096 IPs Private subnet CIDR block in us-east-1b 10.0.144.0/20 — 4,096 IPs
4)	Check the network preview in the right pane.	Mouse over to vpc-testing-igw in Network connection (3). The **Internet Gateway** of vpc-testing-igw connects with the two subnets in the AZ of us-east-1a and the AZ of us-east-1b. 
5)	Choose **1 per AZ** for NAT gateways and **None** for VPC endpoints. Enable DNS hostnames and resolution.	NAT gateways ($) Info Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway None | In 1 AZ | 1 per AZ VPC endpoints Info Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time. None | S3 Gateway DNS options Info ☑ Enable DNS hostnames ☑ Enable DNS resolution

		Select **Create VPC**. Once the VPC is successfully created, select **View VPC.**
6)	Associate the two private subnets which will be allocated to the RDS database	In the left navigation pane, choose **Subnets**. Click vpc-testing-rtb-private1-us-east-1a.  In the right lower pane, choose **Route table**. Note that **Destination 0.0.0.0/0** is set to **Target nat-xxxxxxxx**. This means that traffic destined for the internet (0.0.0.0/0) will be sent to the NAT Gateway. The NAT Gateway will then forward the traffic to the Internet. Choose the **Edit route table associations** tab, click **Edit route table associations**, select and vpc-testing-subnet-private2-us-east-1b, and select **Save**.
8)	Set up the Security Groups	1. Create a security group for the web server. Choose **Security Groups** in the left navigation pane. Click **Create security group** and configure the 1st security group for the web servers. ○ **Security group name:** Web Security Group ○ **Description:** Allow HTTP access ○ **Select** vpc-testing-vpc as **VPC**. In the **Inbound rules** pane, choose **Add rule** to add SSH, HTTP, HTTPS, and MYSQL/Aurora from Anywhere-IP4 0.0.0.0/0.



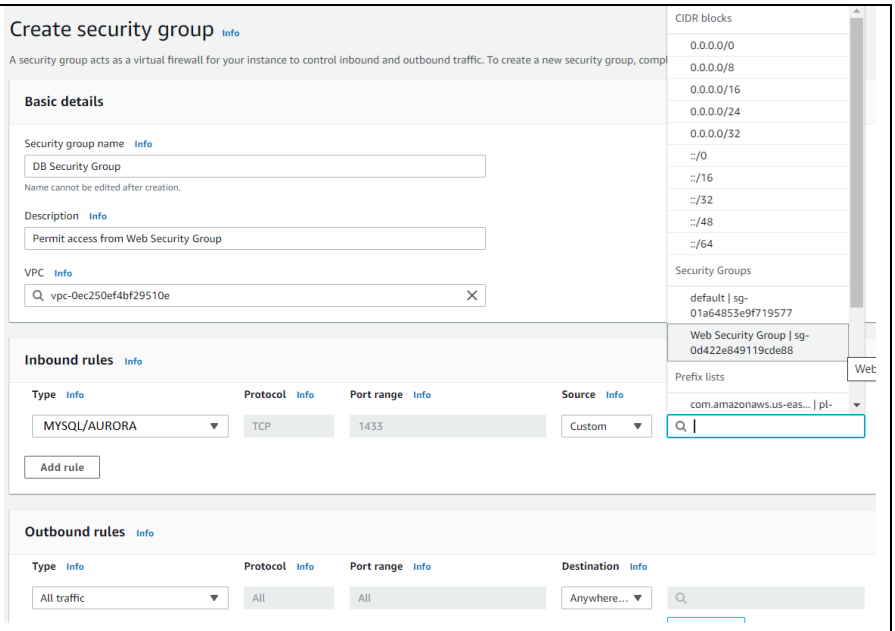
Then, click **Create security group**.

2. Create a security group for the RDS instance.

Click **Create security group** and configure the security group for RDS:

- **Security group name:** DB security group
- **Description:** Allow access from Web Security Group
- **Select** vpc-testing-vpc as **VPC**.

In the **Inbound rules** pane, choose Add rule to add MYSQL/Aurora and select the source as Web Security Group.

		 Then, click **Create security group**.

Task 2: Create an RDS subnet group

In the **Services** search box, search for and select **Aurora and RDS.**

SN	Steps	Instructions
1)	Create and name DB Subnet Group as **DB-Subnet-Group**	In the left navigation pane, choose **Subnet groups**. Choose **Create DB Subnet Group** then configure: ○ **Name:** DB-Subnet-Group ○ **Description:** DB Subnet Group ○ De-select the default VPC and select vpc-testing-vpc as **VPC**. Add two respective AZs and select the two **private** subnets that were created. You can check your previous notes on the CIDR codes for the subnets.

▼ Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

10.0.0.0/20 4,096 IPs

Public subnet CIDR block in us-east-1b

10.0.16.0/20 4,096 IPs

Private subnet CIDR block in us-east-1a

10.0.128.0/20 4,096 IPs

Private subnet CIDR block in us-east-1b

10.0.144.0/20 4,096 IPs

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

DB-Subnet-Group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

DB Subnet Group

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

vpc-testing-vpc (vpc-0ec250ef4bf29510e)

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a ✕ us-east-1b ✕

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

us-east-1a

- [ ] subnet-01b8a1dfbd682344a (10.0.0.0/20)
- [x] subnet-0fd1476d250e04118 (10.0.128.0/20)

us-east-1b

- [x] subnet-02e25955a3f49b85d (10.0.144.0/20)
- [ ] subnet-01fc47eb23454d444 (10.0.16.0/20)

		CIDR block
us-east-1a	subnet-0fd1476d250e04118	10.0.128.0/20
us-east-1b	subnet-02e25955a3f49b85d	10.0.144.0/20

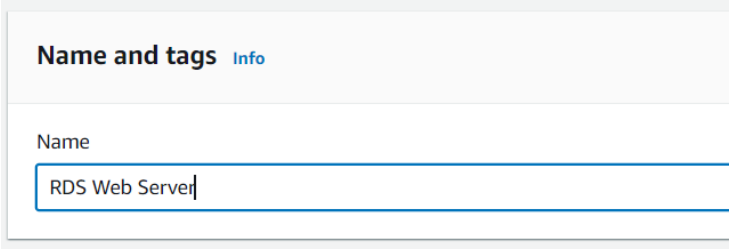
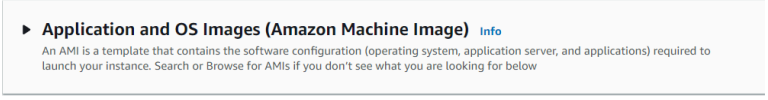
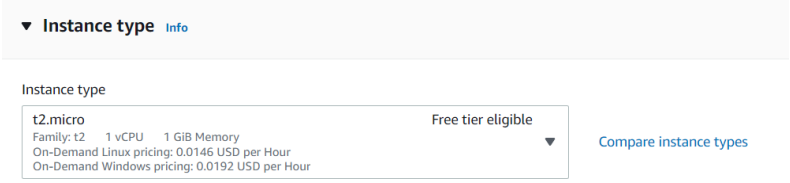
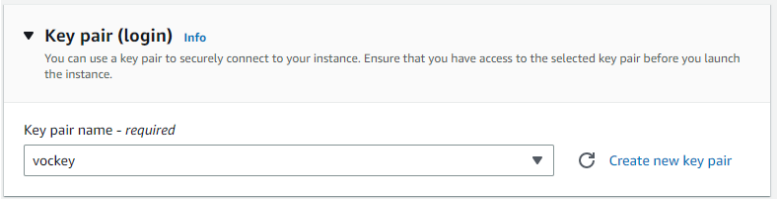
Cancel **Create**

At the bottom of the page, select **Create**. Once the subnet has been created, go to the created subnet group and check the details.

		Add the screenshot to Result 1 in the lab report

Task 3: Launch an EC2 instance

Go to **AWS Management Console**, key in EC2 in the search box of **Services,** select **EC2,** and choose **Launch instances**

SN	Steps	Instructions
1)	Name and Tag: Name the EC2 instance as "**RDS Web Server**"	
2)	Choose the default **AMI**	
3)	Choose the default instance type **t2.micro**	
4)	Choose the default key pair: **vockey**	
5)	Network Settings	Click Edit in the section of **Network settings**, choose vpc-testing-vpc as the VPC and **public subnetwork** from us-east-1a, select **Enable** auto-assign public IP, choose **Select existing security group**, and choose Web Security Group.

<table>
<tr>
<td></td>
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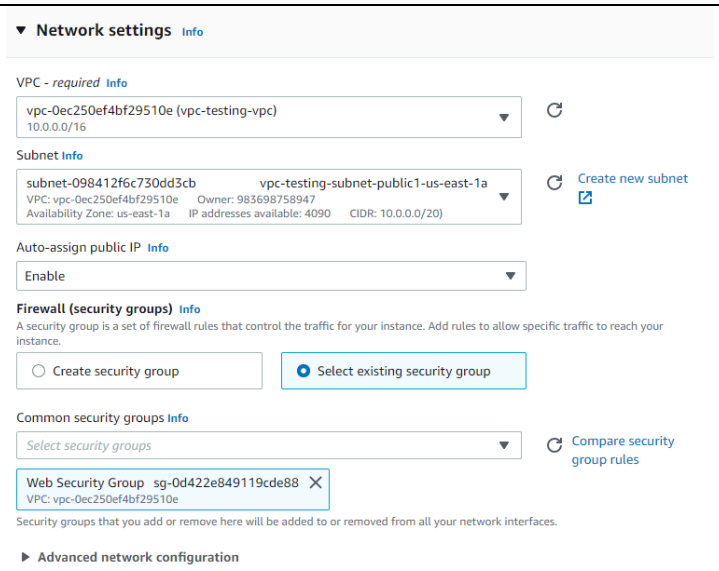
<td>


</td>
</tr>
<tr>
<td>6)</td>
<td>Expand advanced details to configure the user data</td>
<td>
Copy the following scripts into User data.

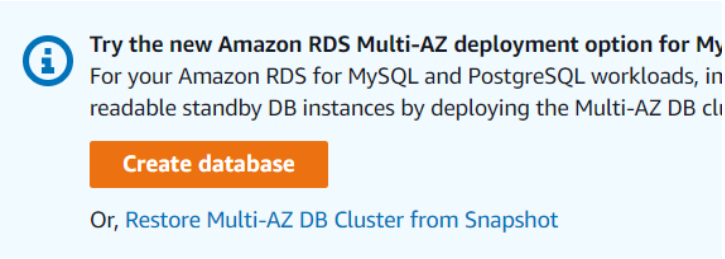
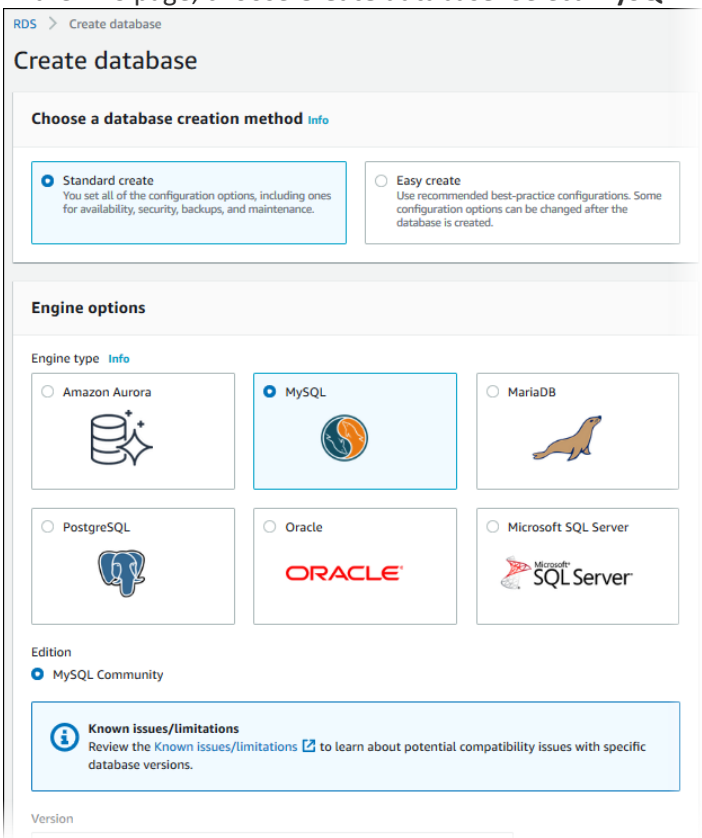
```
#!/bin/bash
# Update package and start Apache web server
sudo yum update -y
sudo yum install httpd -y

# Create a simple index.html file
echo '<html><h1>Hello From Web Server 1!</h1></html>' > a.html
sudo cp a.html /var/www/html/index.html
rm a.html

# Set appropriate permissions
sudo chown apache:apache /var/www/html/index.html
sudo chmod 644 /var/www/html/index.html

#install php
sudo yum install php-cli php-pdo php-fpm php-json php-mysqlnd -y

#install mysql client
sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y
sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023
sudo dnf install mysql-community-client -y

#start the web server
sudo systemctl start httpd
sudo systemctl enable httpd

# Display server status
sudo systemctl status httpd
```

Click **Launch Instance**. Once the instance is launched successfully, choose **View all instances**.
</td>
</tr>
</table>

7)	Check if PHP and MySQL are properly installed on the EC2 instance	1. Select the newly created EC2 instance, click Connect.
		2. In the EC2 Instance command line window, check if php and mysql have been installed properly.
		
		`$php --version`
		
		It should show the version of the installed php.
		
		`$mysql --version`
		
		It should show the version of the installed MySQL client.
		
		If not, please run the following scripts one by one.
		
		#!/bin/bash
		# Update package and start Apache web server
		sudo yum update -y
		sudo yum install httpd -y
		
		# Create a simple index.html file
		echo '<html><h1>Hello From Web Server 1!</h1></html>' > a.html
		sudo cp a.html /var/www/html/index.html
		rm a.html
		
		# Set appropriate permissions
		sudo chown apache:apache /var/www/html/index.html
		sudo chmod 644 /var/www/html/index.html
		
		#install php
		sudo yum install php-cli php-pdo php-fpm php-json php-mysqlnd -y
		
		#install mysql client
		sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
		sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y
		sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023
		sudo dnf install mysql-community-client -y
		
		#start the web server
		sudo systemctl start httpd
		sudo systemctl enable httpd
		
		# Display server status
		sudo systemctl status httpd
		
		Open a web browser and check the URL http://<public_ip_address>

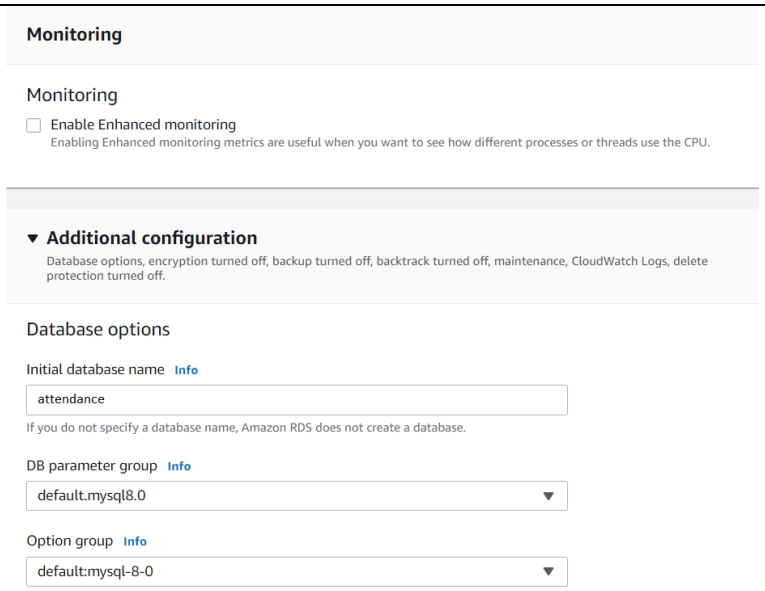
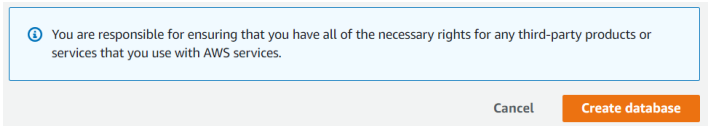
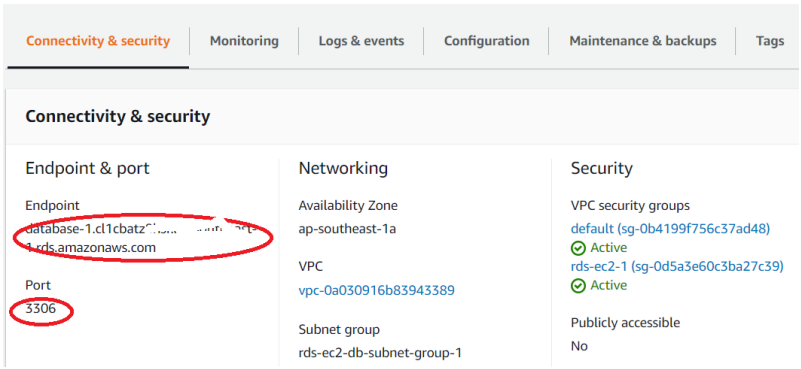
Task 4: Create an RDS database

Go to the search box of **Services** and key in **RDS** and select **Aurora and RDS**.

SN	Steps	Content
1)	Create **RDS database**	Choose **Create database** 
2)	Choose MySQL engine	In the RDS page, choose **Create database.** Select **MySQL.** 
3)	Select Templates: **Free tier**	

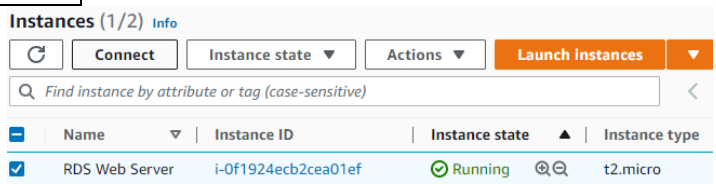
4)	Setting: instance name **database-1** Username: **admin** Set your Password: **testtest**	**Settings** DB cluster identifier Info Enter a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region. database-1 The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen. ▼ Credentials Settings Master username Info Type a login ID for the master user of your DB instance. admin 1 to 32 alphanumeric characters. The first character must be a letter. Credentials management You can use AWS Secrets Manager or manage your master user credentials. ○ Managed in AWS Secrets Manager - *most secure* RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager. ◉ Self managed Create your own password or have RDS create a password that you manage. ☐ Auto generate password Amazon RDS can generate a password for you, or you can specify your own password. Master password Info •••••••• Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @ Confirm master password Info ••••••••
5)	Instance Configuration	Use the default `db.t3.micro` Show instance classes that support Amazon RDS Optimized Writes Info Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost. Include previous generation classes Standard classes (includes m classes) Memory optimized classes (includes r and x classes) Burstable classes (includes t classes) db.t4g.micro 2 vCPUs 1 GiB RAM Network: Up to 2,085 Mbps
7)	Connectivity	Select **Connect to an EC2 compute resource**, and choose `RDS Web Server` as the EC2 instance. Compute resource Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database. ○ Don't connect to an EC2 compute resource Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later. ◉ Connect to an EC2 compute resource Set up a connection to an EC2 compute resource for this database. EC2 Instance Info Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database. i-0f1924ecb2cea01ef RDS Web Server Choose the existing subnet groups: `db-subnet-group`

		DB subnet group Info Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected. Choose existing Choose existing DB subnet group Automatic setup RDS creates a new subnet group for you or reuses an existing subnet group Existing DB subnet groups db-subnet-group Public access Info Yes RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database. No RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database. Choose the existing security group: DB Security Group Public access Info Yes RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database. No RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database. VPC security group (firewall) Info Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic. Choose existing Choose existing VPC security groups Create new Create new VPC security group Additional VPC security group Choose one or more options DB security group × A new VPC security group *rds-ec2-1* will be added to enable connectivity with your compute resource.
8)	Authentication	Use the default Password authentication **Database authentication** Database authentication options Info Password authentication Authenticates using database passwords. Password and IAM database authentication Authenticates using the database password and user credentials through AWS IAM users and roles. Password and Kerberos authentication Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.
9)	Additional Configuration: Set initial database name: **attendance**	After the monitoring section, there is another section “Additional Configuration”. Expand the section to set the initial database name as **attendance**.

		 Uncheck Enable automated backups
10)	Create the database	Choose **Create database** 
11)	Check the status	When the new DB instance is Available, click the DB instance to show its details.
12)	Note the Endpoint	In the Connectivity & security section, get the database endpoint from **Endpoint & Port** of the DB instance.  **Add the screenshot to Result 2 in the lab report**

Task 5: Configure the web server and connect to the RDS instance

Go to search box in **Services**, select **EC2** and go to **instances** in the left pane.

SN	Steps	Content
1)	Connect to the EC2 instance	Select RDS Web Server in the right pane and choose Connect.  In **EC2 Instance Connect** tab, choose **Connect**.

2)	Set permissions for the ec2-user to access Apache server document	1. Change the group ownership of the /var/www directory and its contents to the Apache group.
		`$sudo chown -R ec2-user:apache /var/www`
		2. Change the directory permissions of /var/www and its subdirectories to add group write permissions and set the group ID on subdirectories created in the future.
		`$sudo chmod 2775 /var/www`
		`$find /var/www -type d -exec sudo chmod 2775 {} \;`
		3. Recursively change the permissions for files in the /var/www directory and its subdirectories to add group write permissions.
		`$find /var/www -type f -exec sudo chmod 0664 {} \;`

3)	Connect EC2 to RDS database	1. Log in to MySQL with the RDS endpoint and user name
		`$mysql -h <endpoint> -u <username> -p`
		Replace <endpoint> and <username> with the RDS endpoint and RDS user name e.g., admin.
		Add the screenshot to Result 3 in the lab report
		2. Enter the RDS password and it should show the mysql interaction interface
		`MySQL [(none)]>>_`
		3. Check the database:
		`MySQL [(none)]>Show databases;`

		`+--------------------+`
		`| Database |`
		`+--------------------+`
		`| attendance |`
		`| information_schema |`
		`| mysql |`
		`| performance_schema |`
		`| sys |`
		`+--------------------+`
		`5 rows in set (0.002 sec)`
		
		It should include the `attendance` database that we created before.
		
		4. Exit from MySQL;
		
		`MySQL [(none)]>>exit`

Task 6: Program the web server with PHP

You can follow the steps:

SN	Steps	Instructions

1)	Create a web server config file	1. Under the EC2 instance Connect shell, create a file named `dbinfo.inc` using `vim`.
		
		`$vim dbinfo.inc`
		
		Copy the following content to `dbinfo.inc`
		
		`<?php`
		
		`define('DB_SERVER', '<db_instance_endpoint>');`
		`define('DB_USERNAME', 'admin');`
		`define('DB_PASSWORD', 'testtest');`
		`define('DB_DATABASE', 'attendance');`
		
		`?>`
		
		Replace `<db_instance_endpoint>` with the **RDS endpoint** (e.g., *database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com)*. The RDS endpoint can be copied from the RDS instance.

<table>
<tr><td></td><td></td><td>2. Create a folder under /var/www and copy the file to the folder

<code>$mkdir /var/www/inc</code>
<code>$cp dbinfo.inc /var/www/inc</code></td></tr>
<tr><td>2)</td><td>Create a PHP file for the web page</td><td>1. Create a file named <code>attendance.php</code>

<code>$vim attendance.php</code>

Download <code>attendance.php</code> from LMS x-site and copy the content into <code>attendance.php</code>

2. Copy <code>attendance.php</code> file to <code>/var/www/html</code>

<code>$cp attendance.php /var/www/html</code></td></tr>
<tr><td>3)</td><td>Go to the webpage and add the attendance</td><td>1. Open a web browser and paste the URL below. The public_ip is the IP address or the end point of the EC2 instance.

<code>http://public_ip/attendance.php</code>

2. Add the attendance with STUDENTID and NAME and click Add

<table><tr><td>STUDENTID</td><td>NAME</td></tr><tr><td>123</td><td>Julia</td></tr></table>
Add the screenshot to Result 4 in the lab report

Troubleshooting 1: If the webpage still does not work, connect to RDS Web Server EC2 instance and check if PHP is properly installed.

<code>$php --version</code>

If PHP is properly installed, restart & enable the Apache server.</td></tr>
</table>

		`$sudo systemctl restart httpd`
		`$sudo systemctl enable httpd`
		
		Check if http://<public_ip> shows “Hello From Web Server!”
		
		Troubleshooting 2: if http://<public_ip> works but http://<public_ip>/attendance.php does not, connect to RDS Web Server EC2 instance and check if <db_instance_endpoint> was replaced with the database1 RDS endpoint (e.g., database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com).
		
		`$cat /var/www/inc/dbinfo.inc`
		
		`<?php`
		
		`define('DB_SERVER', '<db_instance_endpoint>');`
		`define('DB_USERNAME', 'admin');`
		`define('DB_PASSWORD', 'testtest');`
		`define('DB_DATABASE', 'attendance');`
		
		`?>`

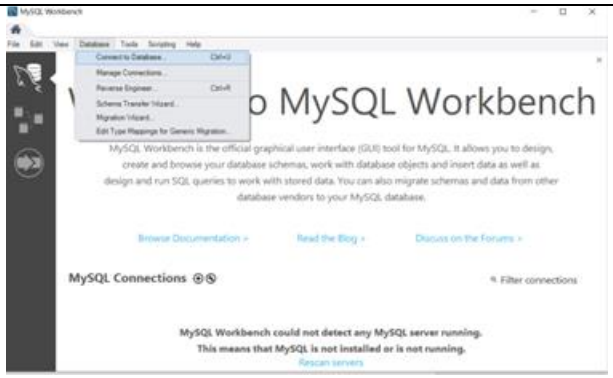
Task 7: Public access of the RDS

SN	Steps	Instructions
1)	Create an EC2 instance in the default VPC	Create an EC2 instance in the default VPC and set the inbound rule allowing SSH traffic. Copy the following scripts into User data. #!/bin/bash # Update package and start Apache web server sudo yum update -y sudo yum install httpd -y # Create a simple index.html file echo '<html><h1>Hello From Web Server 1!</h1></html>' > a.html sudo cp a.html /var/www/html/index.html rm a.html # Set appropriate permissions sudo chown apache:apache /var/www/html/index.html sudo chmod 644 /var/www/html/index.html #install php sudo yum install php-cli php-pdo php-fpm php-json php-mysqlnd -y

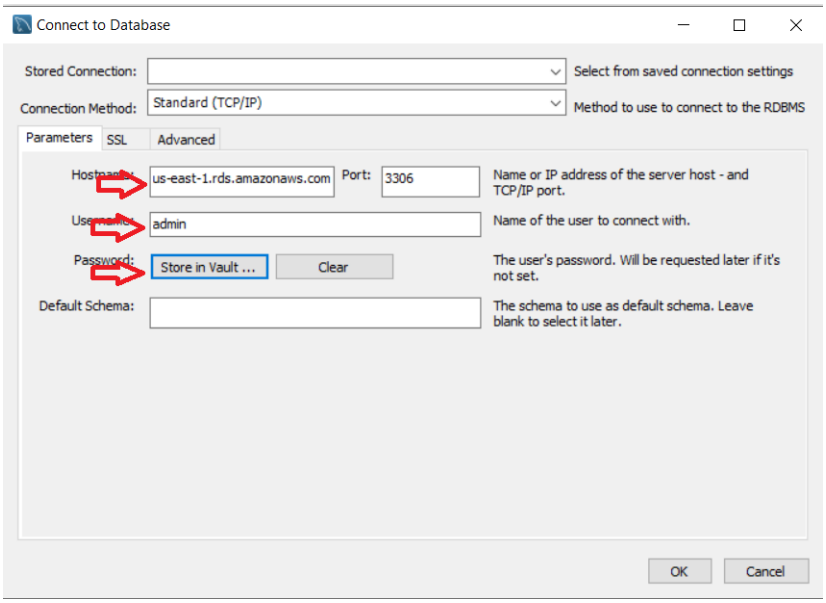
		#install mysql client sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023 sudo dnf install mysql-community-client -y #start the web server sudo systemctl start httpd sudo systemctl enable httpd # Display server status sudo systemctl status httpd Click **Launch Instance**.
2)	SSH into the EC2 Instance	Once the instance is ready, select the instance and click Connect. Connect to the RDS instance using the command line and replace <endpoint> with the RDS endpoint. Key in the RDS password as `testtest`. `$ mysql -h <endpoint> -u admin -p` As the RDS was located in a private subnet, it may **NOT** be able to access from the default vpc, the system may show connection failure. Can you think about why this happens? **Add the screenshot to Result 5 in the lab report**

Students can also install MySQL Workbench 8.0 in their local PC and try remote access to RDS database. **(Optional)**

SN	Steps	Instructions
1)	Connect to Database	1. Open MySQL Workbench, select Connect to Database



2. Fill in the RDS **endpoint** (e.g., *database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com),* and username (ie., admin) and password (ie., testtest).



As the RDS was located in a private subnet, it was not accessible from public network directly. The system may show connection failure.

Task 9 (Optional): Load balancing among multiple RDS web servers

SN	Steps	Instructions
1)	Create a VM image for **RDS Web Server**	Select the RDS Web Server instance, choose **Create image** in Action. Once the image is created, select the image and choose **launch instance from AMI**.

		Launch an instance and name it as **RDS Web Server+1**, select the VPC-testing as the VPC, choose **pub2-us-east-1b** as the subnet, select **Web Security Group** and **ec2-rds-4** as the security group (the same as the security groups in RDS Web Server), and launch the instance.
		
		Launch an instance and name it as **RDS Web Server+2**, select the VPC-testing as the VPC, choose **pub1-us-east-1a** as the subnet, select **Web Security Group** and **ec2-rds-4** as the security group, and launch the instance.
		
		Once the instance is up and running, connect to **RDS Web Server+1** and **RDS Web Server+2**, run the command lines to start the web server.
		
		`$sudo systemctl restart httpd`
		`$sudo systemctl enable httpd`

2)	Create target groups	Select Target Groups in the left navigation pane under Load Balancing section, click **Create target groups**. Name it as **RDSWebServerTargetGroup**, select **vpc-testing** as the VPC, and select RDS Web Server, RDS Web Server +1 and RDS Web Server +2 into the target group.
3)	Create load balancer	Click **Create load balancer,** select **Application Load Balancer,** name it as **RDS-web-lb**, select vpc-testing as VPC, choose **us-east-1a** and its public subnet and **us-east-1b** and its public subnet, select **Web Security Group**, choose **RDSWebServerTargetGroup** to forward to in the Default Action, and click Create load balancer. Once the load balancer is active, get the DNS name of the **RDS-web-lb**, open a web browser, and access the URL http://< RDS-web-lb name/attendance.php>

Task 10: Delete EC2 and RDS

SN	Steps	Instructions
1)	Delete load balancer	Delete **RDS-web-lb** load balancer and delete **RDSWebServerTargetGroup** target group.
2)	Delete RDS	In search box of **Services,** key in **RDS** to go into RDS management console. Select **Databases** in left navigation pane, select database-1. In Action menu, select **Delete**. Uncheck Create final snapshot? and check I acknowledge that upon instance deletion, automated backups, including system snapshots and point-in-time recovery, will no longer be available. Type Delete me to delete the database. Go to **Subnet groups** in the left navigation pane, select subnet groups and click Delete.
3)	Terminate EC2	In search box of **Services,** key in 'EC2' to go into EC2 management console. Select **Instances** in left navigation pane, select RDS Web Server, RDS Web Server +1 and Web Server +2. In Instance state menu, select **Terminate instance**.
4)	Delete VPCs	Select VPCs in the service search box. Go to **NAT gateways** in the left navigation pane, remove the two NAT gateways. Go to **VPC**, select and delete *vpc-testing-vpc*. Go to EC2, release the two **Elastic IPs**.

Write one or two sentences summarizing what you learned from the lab.

Lab complete

Congratulations! You have completed the lab.

Please submit the lab report to the x-Site by end of 31 Mar 2026 (Tuesday).

Thank you!